

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.cluster import KMeans, AgglomerativeClustering
from sklearn.metrics import silhouette_score
```

✓ TASK 1: DATA LOADING

```
df = pd.read_csv("Mall_Customers.csv")

print("First Five Rows")
print(df.head())

print("\nDataset Shape:", df.shape)

print("\nDataset Information")
print(df.info())
```

First Five Rows

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

Dataset Shape: (200, 5)

Dataset Information

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 200 entries, 0 to 199

Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
0	CustomerID	200 non-null	int64
1	Gender	200 non-null	object

```
2   Age                200 non-null    int64
3   Annual Income (k$)  200 non-null    int64
4   Spending Score (1-100) 200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
None
```

✓ TASK 2: DATA PREPROCESSING

```
print("\nMissing Values")
print(df.isnull().sum())

df.fillna(df.mean(numeric_only=True), inplace=True)

encoder = LabelEncoder()
df["Gender"] = encoder.fit_transform(df["Gender"])

X = df.drop("CustomerID", axis=1)

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
Missing Values
CustomerID      0
Gender          0
Age             0
Annual Income (k$) 0
Spending Score (1-100) 0
dtype: int64
```

✓ TASK 3: MODEL BUILDING

```
kmeans = KMeans(
    n_clusters=5,
    random_state=42,
    n_init=10
)
```

```
kmeans_labels = kmeans.fit_predict(X_scaled)

# ----- Agglomerative -----
agglomerative = AgglomerativeClustering(
    n_clusters=5,
    linkage="ward"
)

agg_labels = agglomerative.fit_predict(X_scaled)
```

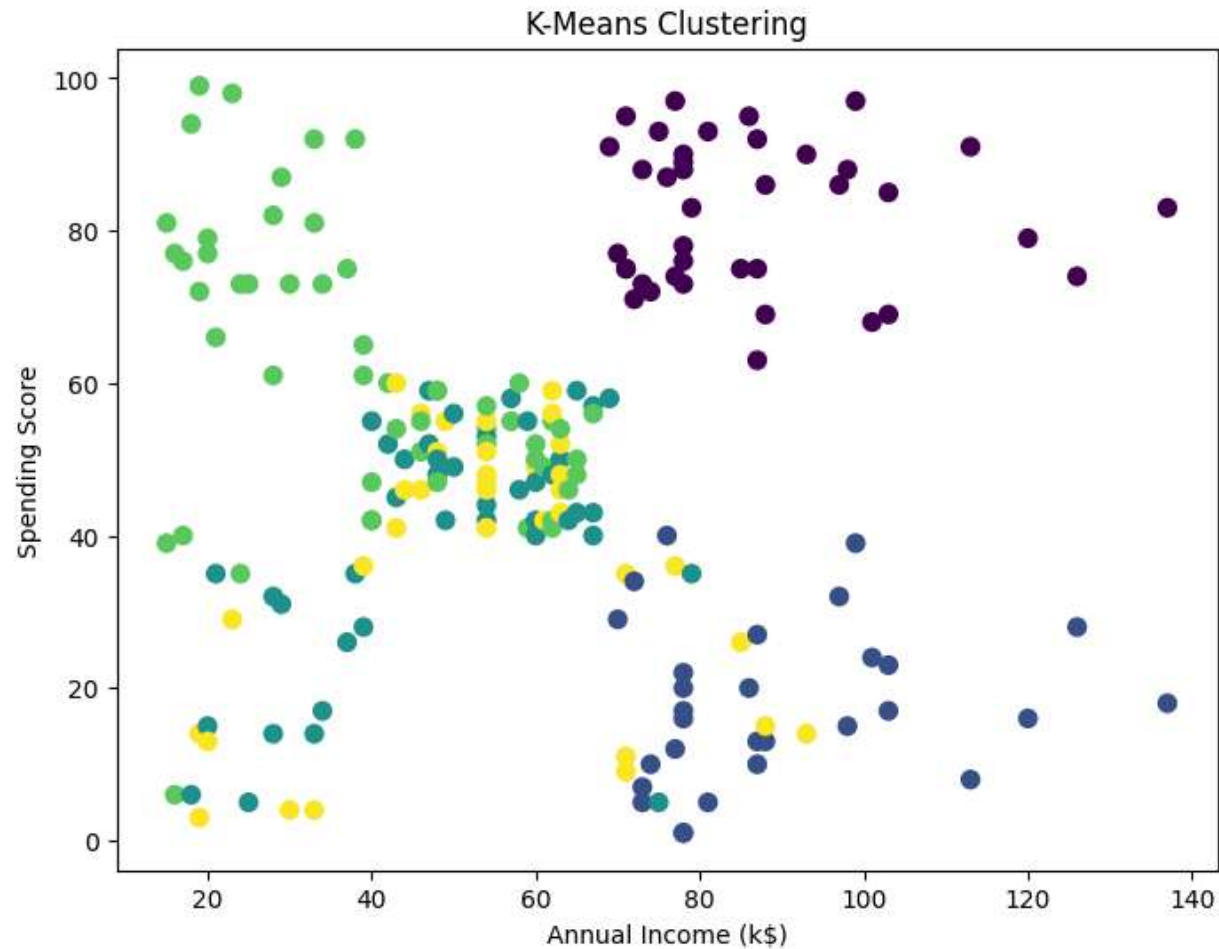
✓ TASK 4: VISUALIZATION

```
features = df[["Annual Income (k$)", "Spending Score (1-100)"]]

plt.figure(figsize=(8,6))

plt.scatter(
    features.iloc[:,0],
    features.iloc[:,1],
    c=kmeans_labels,
    cmap="viridis",
    s=50
)

plt.title("K-Means Clustering")
plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score")
plt.show()
```

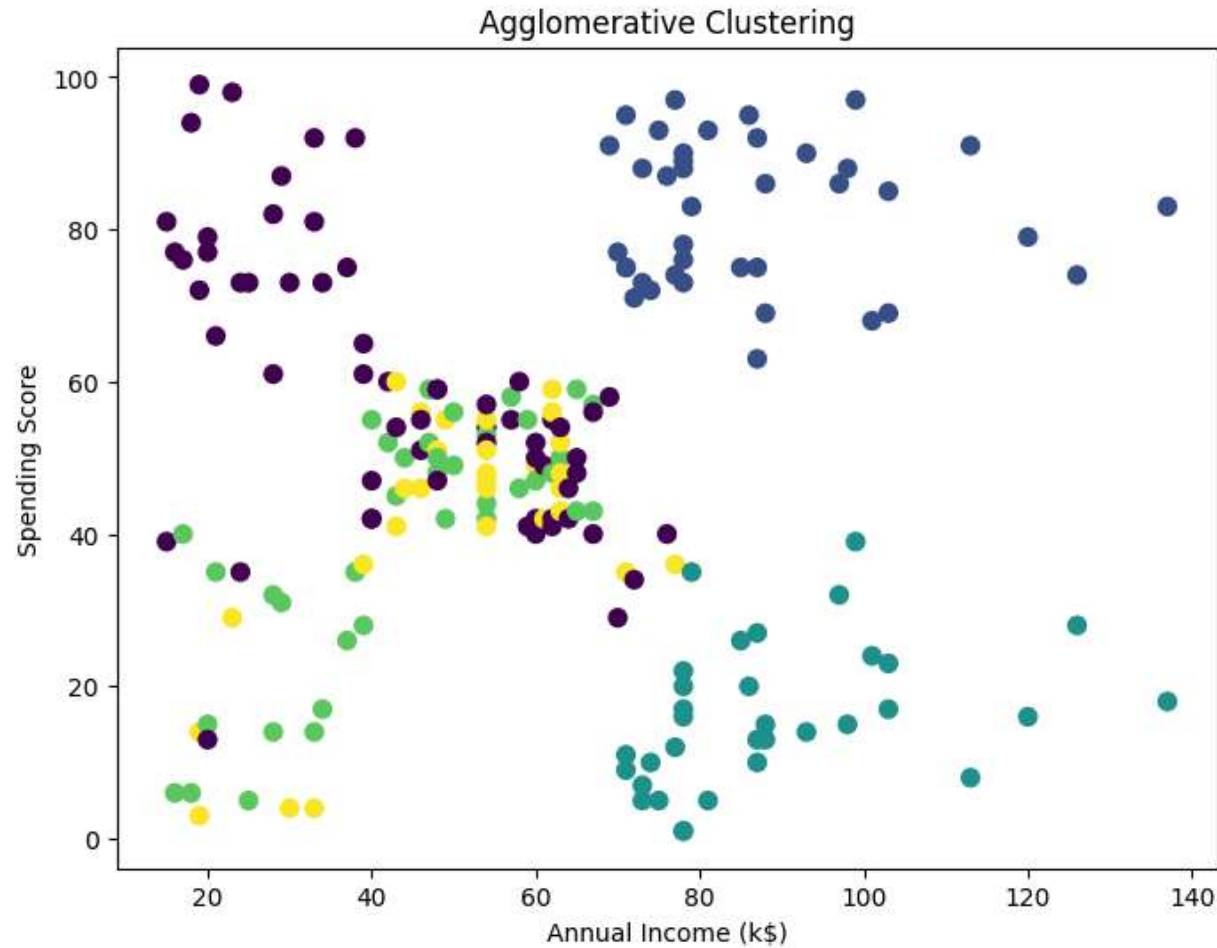


```
plt.figure(figsize=(8,6))

plt.scatter(
    features.iloc[:,0],
    features.iloc[:,1],
    c=agg_labels,
    cmap="viridis",
    s=50
)

plt.title("Agglomerative Clustering")
```

```
plt.xlabel("Annual Income (k$) ")
plt.ylabel("Spending Score")
plt.show()
```



✓ TASK 5: EVALUATION

```
kmeans_score = silhouette_score(X_scaled, kmeans_labels)
agg_score = silhouette_score(X_scaled, agg_labels)

print("\nn=====")
```

```

print("Silhouette Scores")
print("=====")

print(f"K-Means           : {kmeans_score:.4f}")
print(f"Agglomerative      : {agg_score:.4f}")

print("\n=====")
print("Interpretation")
print("=====")

if kmeans_score > agg_score:
    print("K-Means performs better because it has the higher silhouette score.")
elif agg_score > kmeans_score:
    print("Agglomerative Clustering performs better because it has the higher silhouette score.")
else:
    print("Both algorithms perform similarly ")

```

```

=====
Silhouette Scores
=====
K-Means           : 0.3041
Agglomerative     : 0.2870

=====
Interpretation
=====
K-Means performs better because it has the higher silhouette score.

```